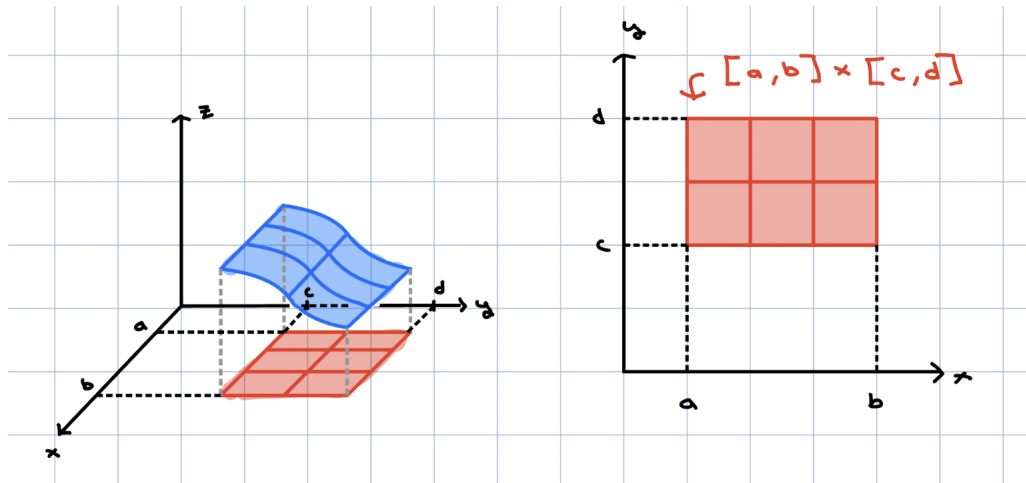


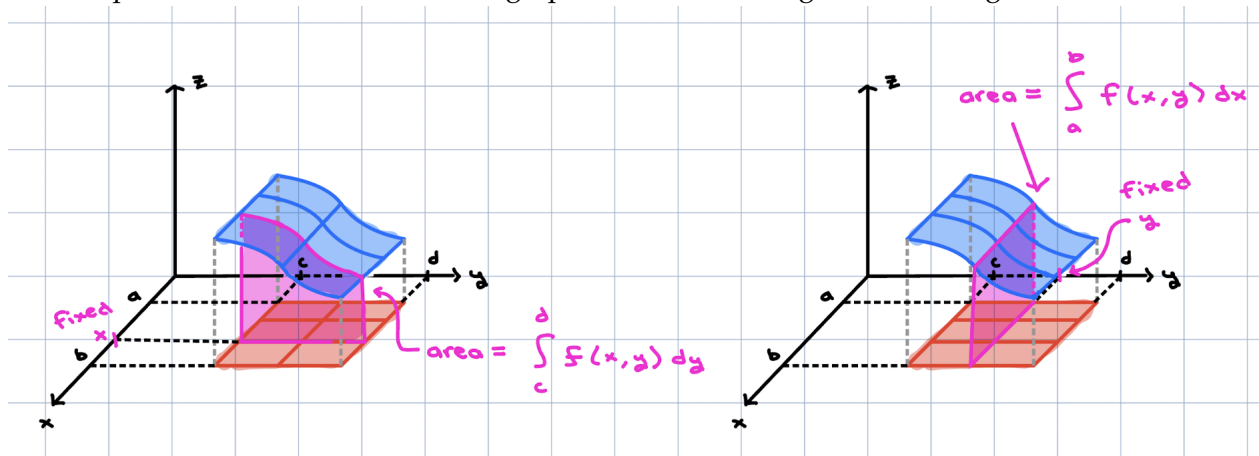
A. Double Integrals Over Rectangular Regions. Let's compute double integrals.

Let $z = f(x, y)$ and consider a **rectangular** region of integration

$D =$



We compute the volume between the graph of f and D using **iterated** integrals.



For a fixed x we can calculate the area of the yz cross-section of this volume.

[area of yz cross-section of this volume] =

For the total volume, we add up these yz cross-section areas for all x .

$$\iint_D f(x, y) \, dA =$$

We could have instead first computed areas of xz cross-sections. This would yield

$$\iint_D f(x, y) \, dA =$$

This exposes the relationship

$dA =$

dA

Example 1. Find the volume of the region bounded above by

$$z = xe^{xy}$$

and below by the rectangle $D = [1, 2] \times [2, 3]$ in the xy -plane.

Do this using an iterated integral in order $dydx$.

Solution. First we set up the iterated integral.

$$\iint_D xe^{xy} \, dA =$$

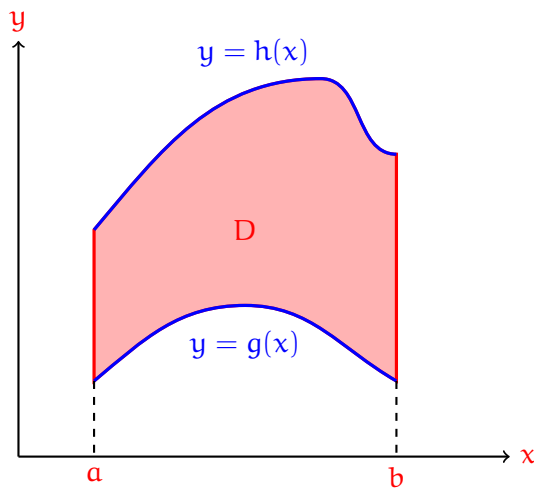
To compute the inner integral we setup a u -substitution.

Now we institute the u -substitution.

B. Double Integrals Over Other Regions. What about non-rectangular regions?

As before consider a graph $z = f(x, y)$ in 3D space.

Let's calculate the volume between the graph of f and region D defined by



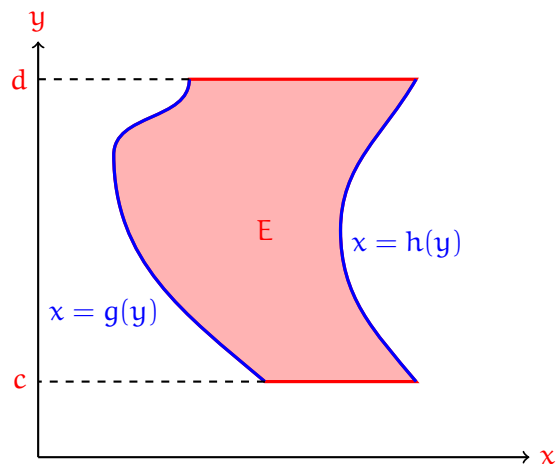
For a fixed x we can calculate the area of the yz cross-section of this volume.

[area of yz cross-section of this volume] =

For the total volume, we add up these yz cross-section areas for all x .

$$\iint_D f(x, y) \, dA =$$

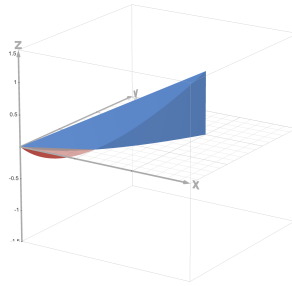
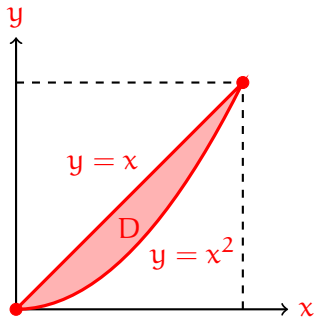
Now let's calculate volume between the graph of f and region E defined by



Using a similar idea, but first calculating xz cross-section areas, we have

$$\iint_E f(x, y) \, dA =$$

Example 2. Let D be the region bounded by $y = x$ and $y = x^2$ and find:



Calculate $\iint_D 2y - x \, dA$ in both orders $dydx$ and $dx dy$.

Solution. Let's mark the points of intersection of $y = x$ and $y = x^2$.

In order $dydx$ we need $\text{function}(x)$ bounds on y and need constant bounds on x .

We now compute the integral

$$\iint_D 2y - x \, dA =$$

In order $dx dy$ we need $\text{function}(y)$ bounds on x and need constant bounds on y .

First we observe that in our region of integration D we have

$$y = x \implies$$

$$y = x^2 \implies$$

Thus we set up bounds.

We now compute the integral

$$\iint_D 2y - x \, dA =$$